Lab 3 - Software Engineering

System Requirements: P2P Shopping Assistant

1. Problem Description

The **P2P Shopping Assistant** is a mobile and web application designed to eliminate the inefficiency of navigating large retail spaces. While standard applications manage lists, they lack the spatial intelligence to guide users to specific items.

This platform leverages decentralized, crowdsourced data to build a dynamic, high-precision map of product locations. By correlating the exact moment a user checks an item off their list with their concurrent **GPS** or **Wi-Fi RTT** coordinates, the system creates a "Peer-to-Peer" information network. This allows the application to suggest the best stores based on item availability and generate optimal in-store routes using a **Traveling Salesperson Problem (TSP)** algorithm.

2. Main Actors and Roles

- **Shopper:** Logs in, creates personal or shared shopping lists, navigates the store using the generated map, and provides location data by marking items as complete.
- **Collaborator:** A secondary user (e.g., family member) who can view and edit shared lists in real-time, allowing for multi-agent shopping.
- **Location Service (External System):** Provides the base mapping data and fine-grained indoor positioning via GPS and Wi-Fi RTT.
- **Retailer Database (External System):** Supplies product metadata such as brand names, standard prices, and images to enrich the shopping list.

3. Main usage for each actor (Use Cases)

Shopper:

- **Create and Manage Shopping List:** The user sets up a list, adds items, and sets parameters like "recurrent" for weekly groceries.
- **Navigate Optimized Route:** The system uses the dynamic map to reorder the shopper's list into a path that minimizes walking distance.
- **Log Item Location:** When an item is checked off, the system automatically sends the location metadata to the server to update the store's "heat map".
- **Store Recommendation:** The user inputs their list, and the system recommends the local shop with the highest probability of having all items in stock based on recent user data.

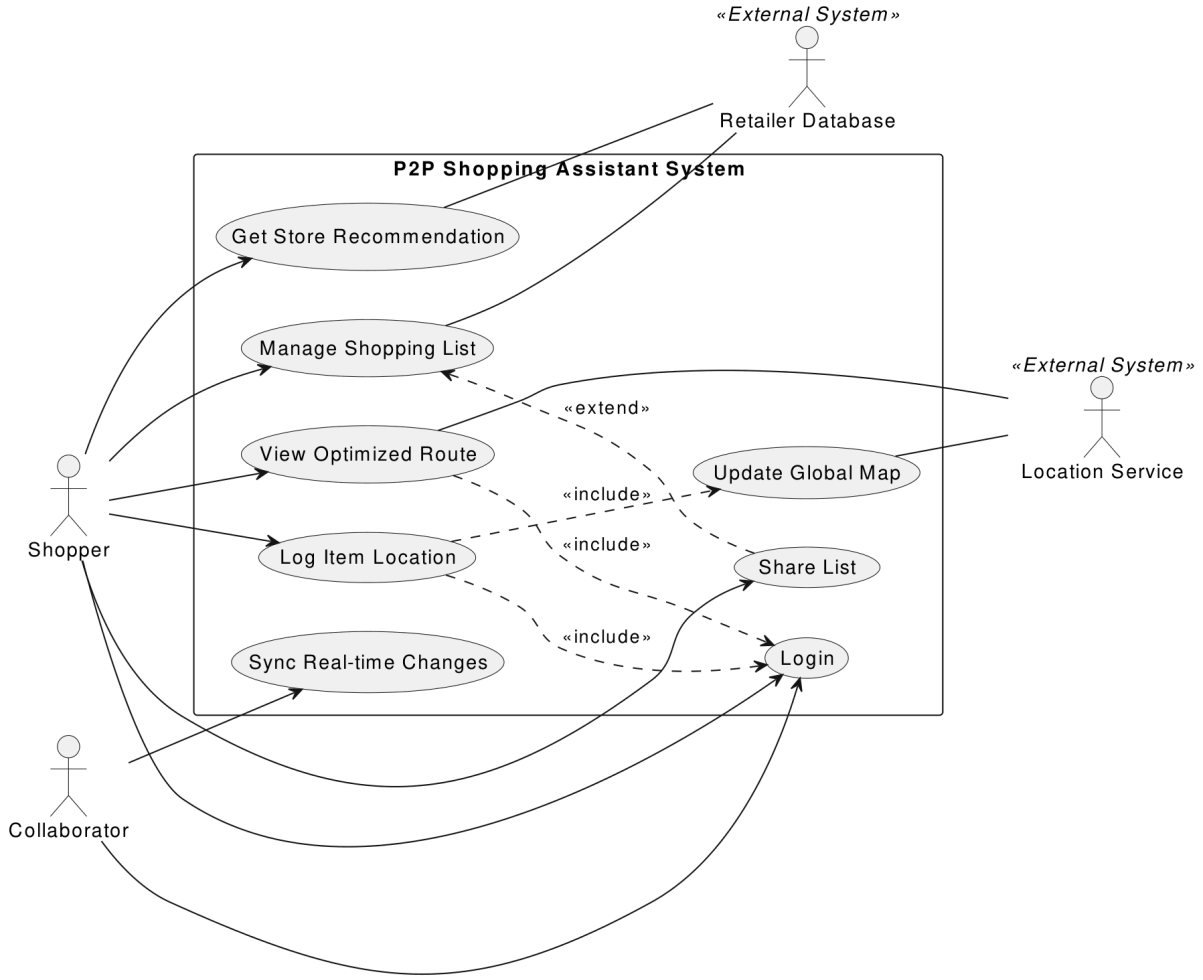
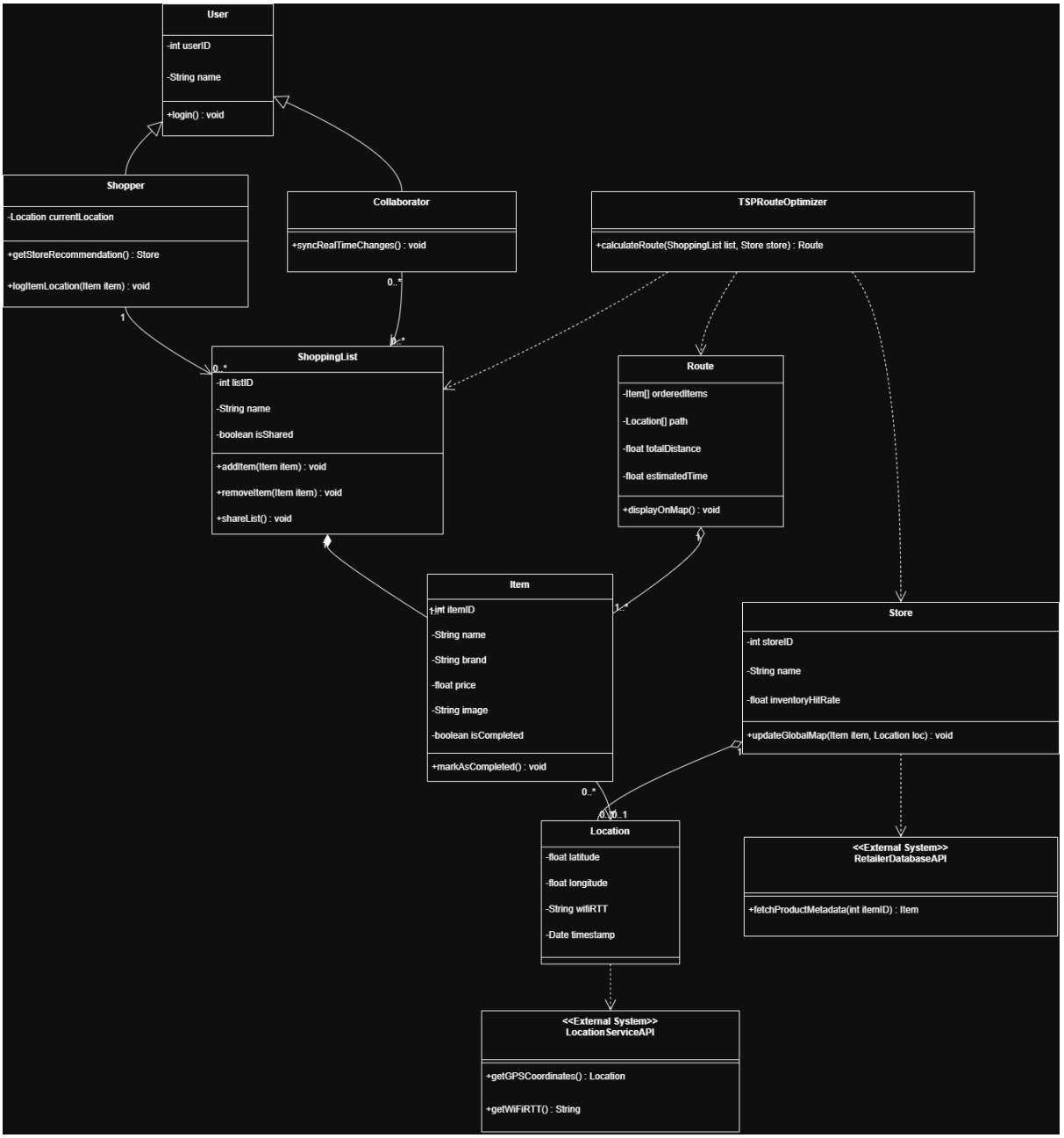
Collaborator:

- **Real-Time List Synchronization:** Multiple users edit the same list; when a collaborator marks an item as "found," it is instantly updated for the Shopper.

Location Service & Retailer (External):

- **Provide Positioning Data:** Supplies the high-accuracy coordinates required for in-store navigation.
- **Validate Product Metadata:** Ensures that the items added by the user match existing product titles and categories.

Use Case Diagram :



Class Diagram :